

Phase 12 Implementation Report — Returns & Refunds

1. Status

Phase 12 complete

2. Exact Verification

- **Typecheck:** `npx tsc --noEmit` — **0 errors**
- **Lint:** `npm run lint` — **0 errors, 0 warnings**
- **Build:** `npm run build` — **PASS**
- **Test Suites:** **94 passed, 94 total**
- **Tests:** **689 passed, 689 total**
- **Failures:** **0 failures**

3. Requirement Matrix

Requirement	Description	Status	Verification Tests
RET-001	Item-level customer return request creation with customer ownership and order validation	COMPLETE	<code>return-api.test.ts</code> , <code>returns-schema.test.ts</code>
RET-002	Server-side return eligibility evaluation preventing duplicate active returns and quantity over-requests	COMPLETE	<code>return-api.test.ts</code> , <code>return-concurrency.test.ts</code>
RET-003	Admin return review (approve/reject) with status transition guards, admin notes, and reviewer tracking	COMPLETE	<code>return-api.test.ts</code> , <code>return-concurrency.test.ts</code>
RET-004	Atomic return approval, refund initiation, and inventory restoration within a single MongoDB transaction	COMPLETE	<code>return-api.test.ts</code> , <code>return-concurrency.test.ts</code>
REF-001	Authoritative manual refund records with non-negative integer minor units (<code>amountMinor</code> , <code>currency "EGP"</code>)	COMPLETE	<code>return-api.test.ts</code> , <code>returns-schema.test.ts</code>
REF-002	Manual refund initiation strictly linked to approved return requests with uniqueness enforcement	COMPLETE	<code>return-api.test.ts</code> , <code>return-concurrency.test.ts</code>
REF-003	Admin manual refund completion with attempt reference, transition to <code>refund_completed</code> , and atomic commit	COMPLETE	<code>return-api.test.ts</code> , <code>return-concurrency.test.ts</code>

4. Files Changed & Created

Created Files

- docs/PHASE-12-RETURNS-REFUNDS.md
- src/database/migrations/scripts/20260928_010_returns_refunds.migration.ts
- src/modules/returns/types/returns.types.ts
- src/modules/returns/models/return-request.model.ts
- src/modules/returns/models/refund.model.ts
- src/modules/returns/repositories/return-request.repository.ts
- src/modules/returns/repositories/refund.repository.ts
- src/modules/returns/schemas/returns.schema.ts
- src/modules/returns/utils/returns.utils.ts
- src/modules/returns/utils/returns.projection.ts
- src/modules/returns/services/returns.service.ts
- src/modules/returns/controllers/returns.controller.ts
- src/modules/returns/routes/returns.routes.ts
- src/modules/returns/index.ts
- tests/unit/returns-schema.test.ts
- tests/unit/returns-refunds-migration.test.ts
- tests/api/return-api.test.ts
- tests/integration/return-concurrency.test.ts

Modified Files

- src/common/errors/errorCodes.ts (Added Phase 12 error codes)
- src/database/migrations/index.ts (Registered Phase 12 migration script 010)
- src/modules/inventory/repositories/inventory.repository.ts (Added atomic product & variant stock restoration)
- src/modules/inventory/services/inventory.service.ts (Added restoreReturnedStock ledger restoration)
- src/app.ts (Mounted customer & admin return/refund routes)

5. Migration

- **Filename:** src/database/migrations/scripts/20260928_010_returns_refunds.migration.ts
- **Migration ID:** 20260928_010_returns_refunds
- **Collections Managed:** returnRequests , refunds
- **Indexes Created:**
 - returnRequests :
 - idx_return_requests_reference_unique ({ reference: 1 } , unique)
 - idx_return_requests_order_created ({ orderId: 1, createdAt: -1 })
 - idx_return_requests_customer_created ({ customerId: 1, createdAt: -1 })
 - idx_return_requests_status_created ({ status: 1, createdAt: -1 })
 - refunds :
 - idx_refunds_return_unique ({ returnRequestId: 1 } , unique)
 - idx_refunds_order_created ({ orderId: 1, createdAt: -1 })
 - idx_refunds_customer_created ({ customerId: 1, createdAt: -1 })
 - idx_refunds_status_created ({ status: 1, createdAt: -1 })

6. API Routes

All routes mounted on `/api/v1` :

Customer Routes

- `POST /api/v1/orders/:orderReference/returns` : Submit item-level return request for a customer's delivered order.
- `GET /api/v1/returns` : List paginated return requests belonging to the authenticated customer.
- `GET /api/v1/returns/:reference` : Get detail of customer's own return request (sanitized safe projection).

Admin Routes

- `GET /api/v1/admin/returns` : List paginated return requests across all customers with status filtering.
- `GET /api/v1/admin/returns/:reference` : Get full return request details including administrative notes and audit references.
- `POST /api/v1/admin/returns/:reference/approve` : Atomically approve return, initiate manual refund record, and restock returned products in the inventory ledger.
- `POST /api/v1/admin/returns/:reference/reject` : Reject return request with admin review notes.
- `GET /api/v1/admin/refunds` : List paginated manual refund records with status filtering.
- `GET /api/v1/admin/refunds/:id` : Get manual refund record detail.
- `POST /api/v1/admin/refunds/:id/complete` : Complete a manual refund record with internal attempt reference.

7. New Tests Added

Test Suite File	Tests	Focus Area
tests/unit/returns-schema.test.ts	8	Zod input schema validation, positive quantity constraints, reason enums, stripping forged client fields
tests/unit/returns-refunds-migration.test.ts	3	Migration execution, idempotent repeated execution, non-destructive <code>down()</code>
tests/api/return-api.test.ts	9	Customer return submission, IDOR ownership isolation, admin approve/reject, manual refund completion, safe customer projection
tests/integration/return-concurrency.test.ts	3	RETURN-RACE-01 (concurrent approval), REFUND-RACE-01 (concurrent refund completion), RETURN-RACE-02 (concurrent return creation)
Total New Tests	23	

8. Open Decisions Preserved

- **OD-15 (Category-specific return periods / rules):** Preserved as open and configuration-driven. No arbitrary hardcoded day limits (such as 7, 14, or 30 days) or hardcoded non-returnable category exclusions were invented.

- **OD-16 (Refund processing SLA):** Preserved as uncommitted. The manual refund status reflects verifiable operational state without promising artificial turnaround windows or automatic timeouts.

9. Regression Status

- All Phase 0–11 suites were re-executed alongside Phase 12.
- **94 suites / 689 tests passed / 0 failures.** Zero regressions across authentication, RBAC, users, categories, products, inventory, carts, orders, payments, shipping, coupons, services, and quotations.

10. Scope Boundary

No Phase 13 Notifications/Realtime delivery work was implemented.

No Phase 14 Outbox worker/jobs/cron work was implemented.

No Phase 15 Reports work was implemented.

No Phase 16 Full Testing phase work was implemented.

No Phase 17 Security Hardening phase work was implemented.

No Phase 18 Deployment work was implemented.

11. Final Stop Condition

Phase 12 is fully implemented, verified, documented, and tested. Execution has stopped at the boundary. Ready for Phase 13 upon instruction.